

KAUMUDI DEGEKAR GULBARGA

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Applied ML and data professional combining statistical analysis, machine learning, and engineering to build robust decision-support systems.

EDUCATION

MS, Software Engineering (<i>Thesis, ML focus</i>) Arizona State University, AZ, USA	Jan 2024 – Dec 2025 4/4 GPA
B. Tech, Computer Science & Engineering G. Narayanamma Institute of Technology & Science, Hyd, IN	Aug 2016 – May 2020 3.3/4 GPA

WORK EXPERIENCE

Software Engineer – Data Analytics, 7 Seas Entertainment	Jun 2021 – Dec 2023
- Engineered reusable SQL pipelines aggregating gameplay, onboarding, and purchase events for analytics. - Analyzed conversion funnels and user behavior; insights drove product changes improving conversion by 12 - Built Tableau dashboards for funnel, retention, and monetization analysis supporting data-driven decisions. - Designed analytics instrumentation enabling reliable revenue reporting and behavioral tracking.	
Software Engineer – Full Stack, Micron Technology	Jul 2020 – May 2021
- Built RESTful APIs and data integration pipelines for fabrication and quality analytics using Angular/JavaScript. - Created Tableau dashboards surfacing production trends, anomalies, and quality metrics for engineering teams.	

RESEARCH & APPLIED ML PROJECTS

Classifying Cell-Cycle Phase from scHi-C Data with Statistical Features and INRs <i>International Conference on Information Systems (IADIS IS), 2026</i>	Fall 2025
- Developed novel approach combining statistical features and per-cell INR embeddings for cell-cycle classification. - Achieved 89% accuracy via ensemble learning, preserving cell-level variability without data imputation.	
Siamese Network for Facial Recognition	Spring 2025
- Implemented a Siamese model with VGG16 and contrastive loss for face-pair verification. - Reached near-perfect accuracy on the AT&T dataset and analyzed model behavior in low-data settings.	
Scalable Implicit Full Waveform Inversion for Velocity Reconstruction	Spring 2025
- Modeled velocity maps as continuous functions using implicit neural representations for high-res reconstruction. - Trained and scaled FWI workflow on HPC cluster using Slurm batch jobs to process 100+ velocity samples.	
Smart Habitat Ranking for Urban Relocation	Fall 2024
- Built a locality recommendation system using knowledge-graph modeling and Semantic Web tools. - Combined multiple datasets into an ontology to compute habitat scores via graph-based reasoning.	

CO-CURRICULAR EXPERIENCE

Graduate Teaching Assistant, Arizona State University	Jan 2025 – Dec 2025
Held office hours, graded assignments, provided individual feedback and personalized support (SER 502).	
Apprenticeship Fellow, Apple Maps Hyd, IN	Dec 2019 – Apr 2020
Designed database schemas, validation, retrieval routines and SQL/JPA logic with expert mentorship.	

TECHNICAL SKILLS

Languages:	Python, SQL, Java, C, Bash; SPARQL, RDF, OWL, SWRL
ML & DS:	Classification, Regression, Clustering; Feature Engineering, Model Evaluation; Pandas, NumPy, scikit-learn, statsmodels, PyTorch, Matplotlib, Seaborn, SHAP, NLP
Tools:	Tableau, MySQL, GraphDB, Jupyter, Linux, Slurm (HPC), Docker, AWS, Git